1. Motivation and significance [make it less narrow else reviewers!!]

In recent years, researchers across disciplines have begun investigating the body and surface touches that infants produce during bouts of spontaneous limb movement. These spontaneous touches are thought to be related to the formation of a somatosensory representation of the body (Corbetta et al., 2014; Thomas et la., 2015; DiMercurio et al., 2018, Khoury et al., 2022, Stupperich et al., 2024). This behavior has important implications for understanding both typical motor development (REF) and atypical development (REF), and it has been proposed as a source of inspiration for robotics and engineering (REF). Despite this growing interest, quantifying these behaviors remains difficult. Although computer vision techniques such as pose estimation (REF) have advanced considerably, they still struggle to detect touch events reliably. Infants' limbs frequently occlude one another, contact is hard to infer from 2D keypoints or even when 3D is available, and current models are trained mostly on adult bodies (REF). As a result, developmental researchers still largely rely on manual annotation.

Unfortunately for this task and ones that share its characteristics (i.e., small time locked behaviors) annotation tools commonly used in the field are poorly suited. Software such as Datavyu (REF) is highly flexible: users define their own coding schemes and can add textual notes describing each event. This makes it well suited to high-level behaviors or complex social interactions, but cumbersome for brief, highly repetitive events. BORIS (REF) offers similar flexibility while allowing events to be coded rapidly through keyboard shortcuts. However, it is still unfit for multiple types of small scale events for two reasons. First, in our example, touches can occur at many different body locations, and coding each location as a separate category quickly becomes tiring for the user. Second, individual touches often last well under a second, which makes real-time coding unreliable and requires frame-by-frame precision.

In this article, we discuss a new software called TinyTouch, a software designed for coding rapid, recurring events and referencing specific points in space.

2. Software description

2.1 General

TinyTouch is an open-source manual annotation tool for coding brief, recurring sensorimotor events from video. It was developed to study spontaneous touches that infants make to their own bodies and to surrounding surfaces, but it can be used for any behavior where both the timing and the location of an event matter. Unlike annotation software that relies on predefined category codes, TinyTouch records each event directly on a map of the body or surface: a left click marks the onset and location of a touch, and a right click marks its offset. Each event is thus captured with frame-level timing and its spatial location in a single action. The result is a frame-by-frame dataset describing the spatio-temporal organization of touch, ready for quantitative analysis.

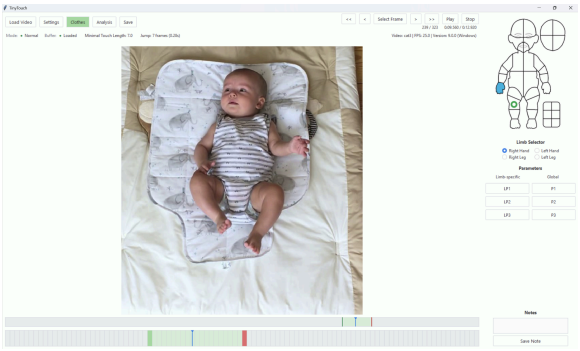
2.2 Software architecture

[Lukas -will do that part as soon as possible]

1 figure (schematic)

2.3 Software functionalities

The user interface is voluntarily simple and stripped from flying or mobile windows to make the coding as comfortable as possible. Functionalities of the software are all focused on making the experience as easygoing and comfortable as possible as we've been reported that labeling sessions can last several hours (see section XXX).



Description of the image here potentially explain briefly that the location map (template) on the upmost corner can be changed in the settings so that we don't have to tell it again or at least to describe it as a Baby map !

2.3.1 Spatial annotation of touches

The locations of events are coded with a simple left click of the mouse on a location map (upper left corner of image XXX). This location map can be adapted and changed by the user according to their needs. In our example case (focused on the labeling of sensorimotor events) we show the body template of an infant with sections of different body locations and surfaces. However the user is free to change it for a more neutral location map. Thus, the user does not have to manually type the code - one just needs to click, which will be automatically registered as an event in the output file (see section XXX).

A clothing section is available for use [POTENTIALLY RENAME THAT, CHECK WITH LUKAS]. It allows the user to select which area of the body map must be treated as special cases. In our example case, researchers can annotate in a few seconds what are the parts of the body that are covered by clothes, which might be particularly relevant to extract when interested in self-touch exploration (REF).

2.3.2 Temporal annotation (onsets and offsets)

The start and stop frames are denoted with the left and right mouse click respectively (green dot for onset and red dot for offset). One should note that (1) those dots can be corrected at any moment during the coding by using the central click of the mouse or by having the mouse pointer in the area of interest and pressing the letter "D" of the keyboard. (2) That several onset and offset can be placed on the location map (either in the same area or in different areas).

When coding, users have to select one of four possible channels which they will code. To take again our case examples, our four channels are the limbs (left and right hands, and left and right

feet). Every channel has its own independent code.

2.3.3 Parameters and notes

Users have access to additional buttons (called parameters) that they can turn on and off depending on their needs and the complexity of their coding scheme. THere are two categories of parameters. (1) button that are independent on the channel in which the users decided to code, that is if they press the button it will stay active at the frame they pressed it regardless of the channel they decide to code in; (2) channel-specific parameters that will only be active in the channel in which they have been pressed.

In many situations users might have the need to describe the situation more precisely, or to add a comment at a particular frame for a later use. They can do that by writing directly in the cell "note" dedicated to this use.

2.3.4 Video navigation

TinyTouch does not require any third-party media player to read the videos. Instead it extracts the frames and lets the user move around frame by frame. Despite using frames directly, it still is able to "play" through the video like any other player (with a slightly reduced speed). This feature protects against any potential dependencies problem, and most importantly it makes the labeling extremely precise, permitting the user to decide without any ambiguity at which frame the event happened (see IMAGE XXXX)

2.3.5 Display and performance settings

[ASK LUKAS for performance TABLE]

2.3.6 Data output

[IMAGE OF THE OUTPUT]

Data is automatically exported when the user closes the software as a .csv file and an .md file that both contain all the information needed for subsequent analysis. They can also be saved by pressing the button "save" during the session.

The output structure can be seen in the section XXX.

2.3.7 Built-in analysis

[IMAGE]

The software can perform descriptive statistics and generate heat maps of the locations labelled of the current file [would that be difficult to pull all files together and produce the same stats?]. These analyses are visualized in an HTML report with touch counts, durations, and charts, plus summary CSV tables, saved in `plots/`. (See section XXX).

In more details see (IMAGE XXX).

Closing remarks

3. Illustrative Examples

4. Impact

4.1 Efficiency

Coding rapid, repetitive behaviors is tedious and time-consuming. Datavyu is one of the most-used software tools for this type of behavior coding. We compared the speed of coding the self and surface touch events in 3-minute video segments using both TinyTouch and Datavyu. This type of touch coding requires

one pass for each hand. Using Datavyu, coding each pass took approximately 1 hour per minute of video; it took a total of 6 hours to code both passes. TinyTouch is much more efficient, taking approximately 15 minutes per minute of video, for a total of 1.5 hours to code both passes. By utilizing a point-click system, TinyTouch allows users to code 3-4x faster. Additionally, TinyTouch does not require a 3rd-party video player, making transitions between video segments smoother. These features make coding small-scale sensorimotor behaviors like touches, finger movements, and twitches more feasible.

This software is also designed with data analysis in mind. While using TinyTouch, data is automatically exported to a csv file that updates each time progress is saved. Additionally, this software can automatically perform descriptive statistics and generate heat maps for locations contacted. This saves valuable time during early stages of data evaluation. Together, these features make identifying and analyzing touches much less time-consuming.

4.2 Applications

TinyTouch is also an open-source tool for frame-by-frame coding with a location map that can be customized to fit different projects' needs. This means that TinyTouch can be used to code various types of infant and adult data. Although TinyTouch was designed for self-touch, its interface is also suitable for coding spatiotemporal events. TinyTouch also has potential applications in validating pose estimation softwares.

like general movements or body twitches (Sokoloff et al., 2021)

Clinical adult data (see paper Asante sent Daniela)

Pose estimation - a way to evaluate quality of the 3D image; features for labelling key points for pose estimation. *image?

5. Conclusion

TinyTouch has been created to meet the growing need for a more precise manual coding software. The main goal of TinyTouch is to provide a more streamlined and user-friendly interface for coding spontaneous touches to the body and surrounding surfaces. When using this tool, event locations are coded with a simple click on a location map of the body and surface; the user does not need to manually type codes. This means that precise locations within a single behavior can be denoted, even if they are only applicable to a few frames. Start and stop frames can be denoted with a left or right mouse click respectively. Parameters can also be coded by turning buttons on and off with a mouse click. Additionally, TinyTouch does not require third party media players. This makes navigating back and forth between frames smoother and more user-friendly.

"Frameline" - bar at the bottom of the video

Datavyu, Boris, Noldus, kinovea -> softwares to code discrete events

OLD

Manual, frame-by-frame coding of behavioral data has been a standard research practice in developmental psychology for decades. However, the study of spontaneous self-touch is new and still largely exploratory. To

understand the role of this behavior in motor development, the precise durations and locations of touches must be coded. These contacts are often shorter than 300ms and move across different areas of the body and surface within a single touch (DiMercurio et al., 2018). This is a more finely-detailed version of frame-by-frame coding than what other softwares were designed for.

Moreover, it has the potential for versatile applications in a wide variety of movement coding fields.